

HERM: HIERARCHICAL EVENT-ROUTED MEMORY FOR CONTEXT-CONDITIONED INTERACTIVE AVATAR GENERATION

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ABSTRACT

Interactive streaming avatars must respond naturally and continuously to conversational partners, yet chunk-wise generators rely mainly on current audio and often produce generic or repetitive motions. Interaction history can help select context-appropriate responses, but retaining full histories is costly, while compressed histories may omit relevant events or fail to guide generation. We propose Hierarchical Event-Routed Memory (HERM), a bounded hierarchy that organizes completed multimodal interactions into events. Before generating each chunk, HERM retrieves relevant trajectories; after generation, it writes the completed chunk back to memory, allowing past interactions to shape subsequent outputs. Experiments show that retrieved history changes subsequent chunks and improves contextual fit, while maintaining comparable visual quality and naturalness.

Index Terms— Streaming avatar generation, multimodal interaction, memory-augmented generation.

1. INTRODUCTION

Interactive streaming avatar generation aims to produce natural, continuous, and responsive animation during conversation. Recent work enables real-time and long-form generation through streaming or autoregressive models [1, 2, 3], while existing interactive systems explicitly model participants’ speaking and listening states using dyadic audio, motion, and conversational context [4, 5, 6, 7, 8]. Yet immediate cues do not specify a unique visual response: the same partner behavior may admit several plausible reactions [9, 10, 11]. Without further context, the generator may repeatedly choose generic motions from this broad response space. Interaction history can reduce this ambiguity by favoring responses that fit both the current situation and the avatar’s established expressive patterns. This motivates history-conditioned generation, where accumulated interaction guides each new full-body response.

We view interaction history as a multimodal event stream, where each response may draw on a different subset of past events. Interactive models already use multimodal context to infer conversational states [6], while query-aware memory methods retrieve different evidence for different queries [12, 13]. The central question is therefore which past events should impact the next response, rather than simply how much history is retained.

Selective history use must remain efficient and reliable as the stream grows. Retaining the full history increases memory and computation with duration [14, 15]. Meanwhile, the history increasingly contains the model’s own outputs, so unfiltered long-term updates may reinforce repetitive or unreliable behavior. This risk is related

to exposure bias and error accumulation in autoregressive video generation [16, 17]. The memory must therefore remain bounded while controlling how generated events become long-term evidence.

To meet these requirements, we propose Hierarchical Event-Routed Memory (HERM) for history-conditioned full-body streaming avatar animation.¹ Unlike memory designed for general video coherence or 3D head-motion style [12, 18], HERM represents interaction history as behavioral events in a bounded hierarchy, initialized from a demonstration and updated with completed chunks. Before generating each chunk, HERM retrieves relevant event trajectories to guide the response. After generation, it updates recent memory and uses calibrated admission control to determine which generated events are retained in long-term memory. This retrieve-generate-update cycle lets past interactions guide future responses while limiting memory growth and reducing unchecked feedback from generated content.

Our contributions are threefold:

- We organize multimodal interaction history into a bounded hierarchy of behavioral events, preserving long-range temporal structure beyond the generator’s local context.
- We develop query-conditioned retrieval and closed-loop memory updates, using distillation and calibrated admission to control how generated events enter long-term memory.
- Controlled experiments show retrieved history changes subsequent chunks, improving contextual appropriateness with comparable visual quality and naturalness.

2. METHOD

HERM adds bounded event memory to a streaming interactive avatar generator, which takes a reference image and streaming audio as input. A demonstration video initializes memory before generation. We describe how completed chunks are organized into events, then how the memory is read and updated during generation, and how these mechanisms are trained and calibrated.

2.1. From Chunks to Bounded Event Memory

To build a bounded event memory from the chunk stream, HERM incrementally organizes completed chunks into events. For each completed chunk, a memory writer extracts aligned visual, audio, and state features, omitting frames repeated across adjacent chunks. It computes event-boundary evidence from novelty relative to the open event, changes in both audio streams, and speaking/listening transitions. After smoothing, a new event starts only when the evidence

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¹Project page: <https://github.com/Wx-Fu/HERM>.

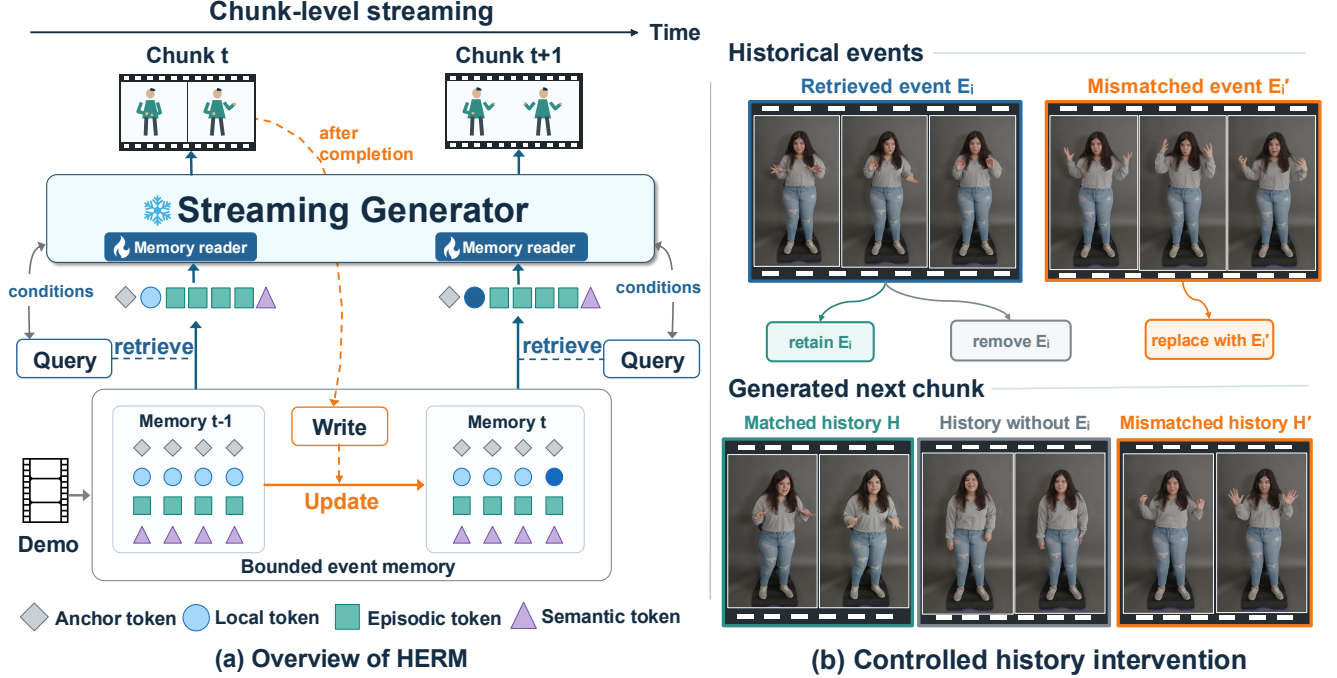


Fig. 1. Bounded event memory and controlled history intervention in HERM. (a) A demonstration initializes memory; current conditions retrieve a fixed-size readout before each chunk, which updates memory after completion. (b) We retain, remove, or replace event E_i with E'_i while holding other history and generation conditions fixed, then compare the next chunks.

exceeds a validation-set threshold; otherwise, the chunk is merged into the open event.

Each event is represented by four field tokens: onset, core, offset, and prototype. Four learned queries with temporal masks extract these fields from the first, middle, last, and full valid windows, respectively. For events spanning multiple chunks, onset and offset are taken from the first and latest valid chunks; core and prototype accumulate intervening content through learned-gated, duration-weighted updates. Local tokens combine pooled visual latents with self/partner audio, state, and temporal/spatial positions. Latent frames shared by adjacent chunks are deduplicated during writing, so each overlapping interval contributes only once.

HERM maintains four token banks: Local stores recent features, Episodic stores event summaries, Semantic stores selected older events, and an immutable Anchor bank preserves reference appearance and scene information. The Episodic bank supports a maximum capacity of $N_e = 48$ events. These banks provide the bounded interaction history used to condition each new chunk.

2.2. Query-Conditioned Retrieval and Updates

As shown in Fig. 1(a), current conditions feed both the streaming generator and a Query path. The Query summarizes self and partner audio, speaking/listening states, and the generator’s motion prefix to form c_k . For each chunk, HERM reads from the existing memory, generates the response, and updates the memory after generation:

$$\begin{aligned} M_k &= R(H_{k-1}, c_k), & \hat{X}_k &= G(C_k; M_k), \\ H_k &= U(H_{k-1}, \hat{X}_k, C_k). \end{aligned} \quad (1)$$

Here, H_{-1} is the initialized memory, C_k denotes the conditions available for chunk k , and c_k summarizes them for retrieval. The

backbone G remains frozen, while the trainable memory readers provide the bounded readout M_k .

Given c_k , the student reader ranks valid Episodic events and selects up to $K = 6$ events. K counts selected events per readout, distinct from the Episodic bank capacity N_e , and B_E denotes the Episodic token allocation. The total readout budget is

$$B = B_A + B_L + B_E + B_S, \quad B_E = 4K, \quad (2)$$

where B_A , B_L , and B_S are the token allocations for Anchor, Local, and Semantic, respectively. The default uses $B = 64$ and $K = 6$; lower budgets reduce the component allocations and K .

At read time, event tokens receive relative-time, duration, and field-role encodings. Readers use event importance, reliability, transition type, and observed/generated source. Relative time is recomputed from stored timestamps, keeping persistent history independent of the current query. Selected Episodic tokens, together with resampled Anchor, Local, and Semantic tokens, form the bounded readout supplied to the generator through gated cross-attention.

Each completed chunk’s non-overlapping content updates Local and the current Episodic event. When an event exceeds Episodic capacity, a retention score favors importance, reliability, usage, and transition coverage while penalizing redundancy and age. An evicted observed event from the demonstration history is consolidated into Semantic through gated merging. An evicted generated event enters Semantic only if it passes calibrated admission; otherwise, it is discarded. Anchor remains immutable, while recent generated chunks remain in Local and the active Episodic event until eviction.

2.3. Memory Distillation and Closed-Loop Adaptation

With the memory construction and runtime read/update policy defined above, we now describe how the student reader is trained and how generated-event admission is calibrated.

Table 1. Main comparison on Seamless Interaction. $\downarrow$ and $\uparrow$ indicate lower and higher is better, respectively. Results are compared only within the same output scope. The best result in each scope is shown in **bold** with a light-blue background.

Method	Scope	Objective metrics						Subjective metrics		
		FID↓	FVD↓	CSIM↑	Sync-C↑	rPCC (%)↓	cMRS↓	Ctx.↑	Nat.↑	ID Cons.↑
<i>Interactive talking-head generation</i>										
Avatar Forcing [19]	Head	374.553	303.523	0.747	1.893	12.099	1.721	3.225	3.125	3.100
DyStream [20]	Head	380.115	323.337	0.817	2.670	10.189	1.257	3.056	2.389	2.528
<i>Body-visible audio-driven animation</i>										
OmniAvatar [21]	Body	90.056	94.617	0.691	3.220	10.792	1.259	3.633	4.100	4.088
LiveAvatar [22]	Body	86.879	99.469	0.651	3.191	9.579	1.102	3.062	2.500	2.812
StableAvatar [23]	Body	83.393	94.711	0.495	1.237	9.669	1.129	3.167	2.792	2.538
G_0 (base generator)	Body	86.684	83.159	0.878	3.694	9.372	1.074	3.577	4.091	4.227
HERM (Ours)	Body	85.497	80.925	0.873	3.767	7.239	0.836	4.023	4.114	4.114

Warm-up trains the memory writer, consolidator, full reader, and confidence head. The head predicts a quality logit from motion jumps, reference drift, and writer event summaries, using Smooth-L1 and ranking supervision based on suffix-normalized latent errors.

After warm-up, an EMA full reader serves as the teacher for the bounded student. The teacher reads all valid Episodic events and other banks under the same storage and eviction constraints; the student router selects up to six events. During training, the Episodic bank capacity N_e varies over $\{8, 16, 24, 32, 48\}$.

The student first uses a learned generic query, then conditions on c_k . Adaptation updates the router, resamplers, and readers, freezing the writer, consolidator, and backbone. We optimize:

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{roll}} \mathcal{L}_{\text{roll}} + \lambda_{\text{distill}} \mathcal{L}_{\text{distill}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}}. \quad (3)$$

Here, $\mathcal{L}_{\text{FM}}$ is the velocity loss; $\mathcal{L}_{\text{roll}}$ matches low-frequency motion changes across rollout boundaries; $\mathcal{L}_{\text{distill}}$ matches student and teacher velocity predictions on the same noisy inputs, excluding the fixed prefix; and $\mathcal{L}_{\text{rel}}$ aligns router probabilities with detached teacher event attention. During student training, the forward pass uses hard top-six selection, while a differentiable assignment over all valid candidates supplies straight-through gradients [24]. Inference uses hard selection alone, with missing event slots masked.

Later training uses autoregressive closed-loop rollouts. Each generated chunk updates Local and Episodic and, under a frozen provisional calibrator, may update mutable Semantic slots before the next chunk. We mix blocked, calibrated, and over-admitted writes, exposing the reader to deployment-like memory states.

After training, the model is frozen and admission is recalibrated using multi-step paired rollouts:

$$H_i(L) = M_{1:L}(\text{admit}_i) - M_{1:L}(\text{block}_i). \quad (4)$$

The paired branches share pre-write memory, conditions, and noise, and differ only in the initial admission decision; both then follow the same policy for L chunks. The lower-is-better score M is a fixed weighted sum of normalized rollout inconsistency, identity drift, boundary discontinuity, and state-response error. An admission is harmful when the score difference exceeds h_{tol} .

Using disjoint held-out splits, we fit Platt scaling from the frozen head’s quality logits to the resulting safety labels and select threshold τ . At inference, events are admitted when their calibrated safety probability exceeds τ .

3. EXPERIMENTS

3.1. Experimental Setup

Dataset. We train and evaluate on Seamless Interaction [25] with participant-disjoint splits, where one participant is the target avatar

and the other the interaction partner. HERM initializes its memory from a non-overlapping target clip.

Baselines and protocol. We compare interactive talking-head methods Avatar Forcing [19] and DyStream [20], and body-visible audio-driven methods OmniAvatar [21], LiveAvatar [22], and StableAvatar [23]. Our backbone is a dyadic interactive talking-video model based on the Wan 14B video diffusion Transformer (DiT); G_0 denotes it without HERM. We follow official settings, matching test identity, audio interval, frame rate, and resolution where supported. Table 1 distinguishes head-only from body-visible results; scores and boldface are compared within each scope.

Metrics. We report FID [26] and FVD [27] for frame and video quality, CSIM from ArcFace embeddings [28] for identity preservation, and Sync-C [29] for lip synchronization. To measure whether the generated response preserves partner–avatar motion coupling, we adapt rPCC [30] to whole-body motion:

$$\text{rPCC} = \frac{1}{D} \sum_{d=1}^D |\rho(v_d^p, v_d^{\text{gt}}) - \rho(v_d^p, v_d^{\text{gen}})|. \quad (5)$$

where v^p , v^{gt} , and v^{gen} are root-aligned, body-scale-normalized pose velocities of the partner, ground-truth target, and generated target. We extract whole-body keypoints with DWPose [31]. Tables report $100 \times \text{rPCC}$ as a percentage; head-only methods use shared facial and head-pose channels and are compared only within scope.

We measure long-range repetition with a GT-calibrated Motion Repetition Score:

$$\text{cMRS} = R + \frac{1}{2} \left| \log \frac{E_g + \epsilon}{E_t + \epsilon} \right| + \frac{1}{2} [C_t - C_g]_+, \quad (6)$$

where R is the mean maximum cosine similarity between normalized 28-frame pose–velocity chunks at least 3.36 s apart; E and C denote motion energy and active-chunk coverage for generated (g) and ground-truth (t) motion. State-specific coverage thresholds are fixed on validation data.

Human Study. Twenty raters scored all model clips in 12 anonymized sets (12 s–2 min) on a five-point scale for contextual appropriateness/partner responsiveness, naturalness/continuity, and identity consistency. The synchronized partner context was shown for contextual assessment, while a reference image or initialization clip was provided for identity assessment.

3.2. Main Comparison

Table 1 compares HERM with existing streaming avatar methods. Since head-only and body-visible systems have different output scopes, we focus the main comparison on body-visible animation. HERM achieves the best FVD, Sync-C, rPCC, cMRS, contextual appropriateness, and naturalness among the body-visible methods.

Table 2. History interventions and ablations. All Δ values use the same 160 samples. Best rPCC and cMRS results are highlighted.

Variant	Δ_{read}	Δ_{shuf} (%)	Δ_{mis} (%)	rPCC (%) $\downarrow$	cMRS $\downarrow$
G_0 (no history)	N/A	N/A	N/A	9.372	1.074
Full-bank reader	2.137	1.265	3.546	11.619	1.386
Fixed query	0.126	2.683	3.934	12.226	1.020
Flat chunk memory	1.683	1.265	4.809	9.579	1.086
w/o online update	0.882	0.232	0.480	9.372	1.072
Ungated admission	1.421	1.265	0.850	11.729	1.051
HERM	1.665	5.258	4.350	7.239	0.836

The largest gains concern interaction behavior: compared with the controlled backbone G_0 , rPCC decreases from 9.372% to 7.239%, cMRS from 1.074 to 0.836, and contextual appropriateness increases from 3.577 to 4.023. These changes indicate closer agreement with ground-truth partner–avatar motion correlations, less repetitive motion, and better contextual fit.

The improvement is not limited to response metrics. HERM reduces FVD from 83.159 to 80.925 and slightly improves FID over G_0 , while maintaining comparable naturalness. Although its CSIM and identity-consistency ratings are slightly lower than G_0 ’s (0.873 vs. 0.878 and 4.114 vs. 4.227), the differences are small relative to the gains in response metrics. HERM improves partner responsiveness and motion diversity without notable visual degradation.

3.3. Does HERM Use Relevant History?

We test whether retrieved history is readable and useful by fixing the inputs and noise. The Anchor-only condition removes retrieved history. The shuffled condition reassigns all four raw Episodic fields while preserving metadata, whereas the content-replaced condition substitutes same-identity events matched by role, duration, and motion energy. Figure 1(b) visualizes these interventions: retaining, removing, or replacing E_i changes the next generated chunk while remaining history and conditions are fixed.

We measure direct output influence with

$$\Delta_{\text{read}} = \left[\frac{1}{|\mathcal{V}|} \sum_{u \in \mathcal{V}} \left(\Delta \tilde{P}_{\text{true}}(u) - \Delta \tilde{P}_{\text{anchor}}(u) \right)^2 \right]^{1/2}, \quad (7)$$

where u indexes valid time, joint, and coordinate entries of the first-differenced, hip-root-aligned, shoulder–hip-scale-normalized DW-Pose coordinates. A larger Δ_{read} indicates stronger history influence, but not necessarily a better response.

We measure content dependence using

$$\begin{aligned} \Delta_{\text{shuf}} &= \text{rPCC}_{\text{shuffled}} - \text{rPCC}_{\text{true}}, \\ \Delta_{\text{mis}} &= \text{rPCC}_{\text{replaced}} - \text{rPCC}_{\text{true}}. \end{aligned} \quad (8)$$

Here, “true” denotes intact-history generation. Positive gaps indicate that intact history yields lower rPCC than the corresponding intervention. All interventions use the same 160 valid samples. We report rPCC and the paired gaps in percentage points, whereas Δ_{read} remains a normalized pose–velocity distance.

Table 2 separates history influence, content dependence, and usefulness. Simply exposing the generator to more history is insufficient: the full-bank reader changes output most but degrades rPCC and cMRS, indicating unselective history use. HERM instead depends on the retrieved event content: shuffling or replacing history raises rPCC from 7.239% to 12.497% and 11.589%, respectively. Flat memory has a comparable replacement gap but worse quality, so content dependence alone is insufficient. Fixed-query retrieval

Table 3. Readout-budget and long-rollout results. K is the number of Episodic events selected per readout; the shaded row is default; Δ Lat. is relative to G_0 .

Setting	K	FVD $\downarrow$	rPCC (%) $\downarrow$	cMRS $\downarrow$	Δ Lat. (ms) $\downarrow$
<i>Readout budget B (tokens)</i>					
16	2	80.067	9.590	0.901	241
32	4	81.945	8.424	0.856	259
64	6	80.925	7.239	0.836	283
128	6	81.159	7.990	0.821	311
<i>Long rollout ($B = 64$, $K = 6$)</i>					
Early	6	79.939	7.320	0.828	283
Middle	6	80.990	7.186	0.831	283
Late	6	82.041	7.211	0.846	286

also fails to adapt memory to the current condition. Removing on-line updates nearly eliminates the intervention gaps, while ungated admission degrades response quality. Together, these results support event structure, condition-guided retrieval, and controlled online updates for responsive, less repetitive generation.

3.4. Efficiency and Long-Stream Stability

Table 3 identifies 64 tokens as a practical operating point. Raising the budget from 16 to 64 tokens raises selected Episodic events per readout from two to six and expands Anchor/Local/Semantic readout, reducing rPCC from 9.590 to 7.239% and cMRS from 0.901 to 0.836, at 42 ms added latency and 1.1% higher FVD. At $K = 6$, expanding the budget to 128 tokens enriches the non-Episodic readout but improves cMRS by only another 1.8%, while rPCC worsens by 10.4% and latency increases by 28 ms. We therefore select 64 tokens as the default: additional bank readout does not reliably improve response quality and may dilute relevant context.

For the 64-token setting, we evaluate five-minute test videos whose rollouts reach memory saturation and undergo repeated event eviction and merging. From Early to Late, relative latency remains nearly constant at 283–286 ms, while rPCC stays within 7.186–7.320% and cMRS within 0.828–0.846, indicating no progressive degradation in the measured interaction behavior. FVD increases from 79.939 to 82.041 (2.6%), suggesting modest visual drift. These results support bounded read cost and stable behavior over the evaluated long-rollout horizon.

4. CONCLUSION

Streaming interactive avatars generate visually coherent, continuous animation with appearance quality, yet often fall back on generic motions. The challenge is not merely producing plausible movement: immediate cues leave several valid choices, while absent interaction history hinders selecting a response suited to the exchange. We introduced HERM to address this problem by organizing completed chunks into bounded multimodal events, retrieving relevant history under current conditions, and controlling how generated events are consolidated into long-term memory. Controlled interventions and ablations show that selected event content changes subsequent chunks, while structured updates and calibrated admission make this influence useful. Experiments show improved contextual appropriateness and partner–avatar motion coupling, less repetition, and comparable appearance quality, with stable read cost and behavior over long rollouts. Overall, HERM shows that selective, controlled interaction memory makes streaming avatar animation more responsive and varied without unbounded history.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study uses the Seamless Interaction dataset, whose creators report obtaining participants' consent for recording, and includes a human evaluation with 20 raters, all of whom provided informed consent. Interactive avatar generation can enable imitation of real people; generated videos should therefore be used only with appropriate authorization and identified as synthetic. Researchers should respect participants' consent and privacy and avoid using recordings or generated outputs to identify individuals or infer sensitive traits.

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